AMAN KUMAR

AI ENGINEER & FULL-STACK DEVELOPER

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Portfolio | GitHub / ReaperXD67 | LinkedIn / Aman Kumar

SUMMARY

AI engineer and former micro1 intern building large language model (LLM) applications and full-stack products. Focus: durable agents, grounded retrieval, Python APIs and TypeScript interfaces.

EXPERIENCE

Project Lead Developer Intern **Jul 2026 - Present**

SIP Organization | Remote

- Lead development of AI-powered WhatsApp onboarding and customer-assistance workflows, including context-aware conversations and response orchestration.
- Integrate AI microservices and backend workflows; coordinate reliability and end-to-end product delivery.

AI Engineer Intern **Aug 2025 - Jul 2026**

micro1 | Remote

- Implemented automatic worker spawning and dynamic task allocation in a self-adaptive AI architecture.
- Built modular cells for inter-cell communication, workload distribution, resource management and monitoring.

Independent AI / Machine Learning Developer **2025 - Present**

Freelance | Remote

- Deliver AI and automation workflows with PyTorch, FastAPI, Dify and n8n across training, evaluation, inference and data-processing pipelines.

PROJECTS

Autonomous Personal Agent | Durable Agent Control Plane **2026**

Python, FastAPI, PostgreSQL, Redis, Docker | Local alpha | Source

- Designed a PostgreSQL transactional outbox and Redis worker queue with leases, heartbeats and bounded retries so interrupted agent tasks can recover.
- Bound human-in-the-loop approvals to exact payloads; rejected stale worker results and checked durable receipts to block replay of recorded external actions.
- Integrated hosted LLM routing with local Qwen fallback; isolated Docker action workers and gated CI on dependency, secret and container-image scans.

AtlasLM | Evidence-Grounded RAG Workbench **2026**

Next.js, TypeScript, Qdrant, Upstash Vector, OpenRouter | Live | Source

- Built PDF, TXT and Markdown ingestion with deterministic chunk IDs, exact deduplication and source metadata carried into inspectable answers.
- Combined dense vector search, BM25, reciprocal rank fusion and reranking; gated weak evidence before generation and audited citations afterward.
- Deployed on Vercel with Upstash Vector and local Qdrant support; added retrieval evaluations, timed pipeline traces and document-scoped semantic caching.

MinePulse / KarixMC | Minecraft Marketplace + Paper Plugin **2026**

Next.js, TypeScript, Java, PostgreSQL, Redis, Prisma, Nginx | Live | Source

- Built and deployed a Next.js marketplace and Java Paper plugin for account linking, server-verified playtime rewards and in-game purchase delivery.
- Designed transactional ledgers and expiring purchase claims; persisted plugin receipts so a lost acknowledgement need not repeat a recorded delivery.
- Authenticated the game/API boundary with HMAC-SHA256 and one-time nonces; deployed app replicas, PostgreSQL and Redis behind Nginx with encrypted backups.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL
AI / ML: LLMs, retrieval-augmented generation (RAG), agent workflows, PyTorch, embeddings, reranking, evaluation
Backend / Data: FastAPI, Next.js, React, REST APIs, PostgreSQL, pgvector, Redis, Qdrant, Prisma, OAuth
Infrastructure: Docker, Linux, Nginx, GitHub Actions, CI/CD, Vercel, VPS, observability, webhooks

EDUCATION & CERTIFICATION

Bachelor of Science in Computer Science **Aug 2024 - Sep 2028**

Scaler School of Technology in collaboration with BITS Pilani | Bengaluru, India

Certified Freelance AI / Machine Learning Developer - micro1, Mar 2026 | Certificate